

XUANFEI REN

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Education

University of Wisconsin-Madison

PhD student, advised by Tengyang Xie, Department of Computer Sciences.

Madison, USA

Aug. 2024 –

University of Science and Technology of China

Bachelor of Science in Mathematics, specialization in Probability and Statistics.

Hefei, China

Aug. 2020 – Jun. 2024

Publication

- [Xuanfei Ren](#), Tengyang Xie.
When Does Trajectory-Level Supervision Permit Efficient Offline Reinforcement Learning?
Submitted to NeurIPS 2026.
- [Xuanfei Ren](#), Allen Nie, Tengyang Xie, Ching-An Cheng.
POLCA: Stochastic Generative Optimization with LLM
In Proc. of the 43rd International Conference on Machine Learning (ICML 2026).
- [Xuanfei Ren](#), Tianyuan Jin, Pan Xu.
Optimal Batched Linear Bandits.
In Proc. of the 41st International Conference on Machine Learning (ICML 2024).

Research Interests

My current research interests span broadly across **AI agents** and **reinforcement learning**, encompassing both theoretical foundations and experimental work.

Research Experience

Outcome-Based Offline Reinforcement Learning

Work with Prof. Tengyang Xie

- Developed a statistical theory for **offline reinforcement learning from trajectory-level supervision**, where datasets provide only scalar outcome labels instead of process-level rewards.
- Proposed **OPAC (Outcome-based Pessimistic Actor-Critic)**, which jointly learns a latent per-step reward model and optimizes policies via pessimistic actor-critic updates.
- Proved a sharp high-probability guarantee of order $\tilde{O}(H^2 \sqrt{C_{\text{sa}}(\pi^*)/n})$ and established a matching lower bound, characterizing the statistical cost of replacing H step-level rewards with one trajectory-level label.
- Extended the framework to **trajectory-level preference feedback** under the Bradley–Terry–Luce model, showing that pairwise comparisons can preserve the same leading horizon and concentrability dependence.
- Introduced a generalized outcome-based RL formulation for nonlinear trajectory objectives, proving an exponential lower bound for all-success objectives and identifying tractable regimes via the structural coefficients $\kappa_\mu(\sigma)$ and $\chi_\mu(\sigma)$.

Stochastic Generative Optimization with LLM

Work with Dr. Ching-An Cheng (Google Research), Dr. Allen Nie (Google Deepmind) and Prof. Tengyang Xie

- Formalized complex system tuning as a **stochastic generative optimization** problem, utilizing LLMs as optimizers to automate iterative refinement through numerical rewards and natural language feedback.
- Engineered a scalable framework featuring a **priority queue-based search strategy** to balance exploration-exploitation, systematically tracking solution histories under noisy evaluation conditions.
- Integrated an **ϵ -Net mechanism** to ensure parameter diversity and an **LLM Summarizer** for meta-learning, significantly reducing redundant trials and improving sample efficiency.
- Demonstrated state-of-the-art performance across diverse benchmarks: **τ -bench** (agent orchestration), **VeriBench** (code translation/Lean4), and **KernelBench** (high-performance CUDA kernels).
- Contributed to **OpenTrace**, an open-source library for end-to-end generative workflow optimization, enhancing its support for robust stochastic behaviors.

Optimal Batched Linear Bandits.

Advisor: Prof. Pan Xu (Department of Computer Science, Duke University.)

Duke University

Aug. 2023 – Jan. 2024

- Devised an algorithm striving for asymptotic and non-asymptotic optimality simultaneously in the **linear bandits regret minimization** setting, an achievement previously unattained.
- Adapted the algorithm into a batched version with **provable least batch complexity**, extending applicability to common real-world problems.
- Confirmed the algorithm's superiority over existing baseline methods through rigorous experimentation, showcasing its practical efficacy in linear bandits problems.

Community Service

Conference Reviewer: ICML, ICLR, RLC, ACL.

Skills

English

Programming: Python, PyTorch, R, LaTeX, MATLAB, Mathematica.